MIL-I-50711
15 December 1970

MILITARY SPECIFICATION

IRON-BASE SINTERED METAL BARS (OIL IMPREGNATED)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers iron base sintered metal powder, oil impregnated bars.

1.2 Classification.

Grade 1 - Iron-carbon base - 0.25 percent max. carbon
Grade 2 - Iron-carbon base - 0.25 to 0.6 percent carbon
Grade 3 - Iron-copper base - 7.0 to 11.0 percent copper
Grade 4 - Iron-copper base - 18.0 to 22.0 percent copper

1.3 Form. Sintered bars shall be furnished in the form of solid bars or cored bars, as specified.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-576 - Box, Wood-Cleated, Veneer Paper Overlaid
PPP-B-585 - Box, Wood-Wirebound
PPP-B-591 - Box, Fiberboard, Wood-Cleated
PPP-B-601 - Box, Wood, Cleated Plywood
PPP-B-621 - Box, Wood-Nailed and Lock Corner
PPP-B-636 - Box, Fiberboard
PPP-B-640 - Box, Fiberboard, Triple Wall

FSC 9510

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MIL-P-116 - Preservation,Methods of
MIL-L-6083 - Lubricating Oil, Instrument, Aircraft, Low Volatility
MIL-L-15016 - Lubricating Oil, General Purpose

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals, Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

B 328 - Density and Interconnected Porosity of Sintered Powder Metal Structural Parts and Oil-Impregnated Bearings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

## 3. REQUIREMENTS

3.1 Specific. Bars shall be made by briquetting and sintering metal powders so as to produce finished parts conforming to the requirements of this specification.



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3.2 Chemical requirements.

3.2.1 The material shall conform to the chemical requirements shown in table I.

Table I. Chemical requirements

| Element | Composition, percent Grade 1 | Grade 2 | Grade 3 | Grade 4 |
|---|---|---|---|---|
| Copper | | | 7.0-11.0 | 18.0-22.0 |
| Iron | 96.25 min. | 95.9 min. | remainder[a] | |
| Total other elements by difference, max. | 3.0 | 3.0 | 3.0 | 3.0 |
| Combined carbon[b] (on basis of iron only) | 0.25 max. | 0.25-0.60 | | |
| Silicon, max. | 0.3 | 0.3 | | |
| Aluminum, max. | 0.2 | 0.2 | | |

[a] Total of iron plus copper shall be 97 percent, min.
[b] The combined carbon may be a metallographic estimate of the carbon in the iron.

3.2.2 Analysis shall be routinely made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated or suspected in amounts greater than the specified limits, further analysis shall be made to determine that the total of other elements is not in excess of the specified limits.

3.2.3 The contractor shall furnish an analysis of each lot showing the percentage of each of the elements specified in table I.

3.3 Physical requirements.

3.3.1 The density of the bars supplied fully impregnated with lubricant conforming to MIL-L-6083 or MIL-L-15016 shall be within the limits specified in table II.

Table II. Density requirements (oil impregnated)

| Grade | Density, g/cm$^3$ Min. | Max. |
|---|---|---|
| 1 and 2 | 5.7 | 6.1 |
| 3 and 4 | 5.8 | 6.2 |



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3.3.2 Oil content. Oil content of this material shall not be less than that designated in table III when determined in accordance with ASTM Method B 328.

Table III. Oil content requirements

| Grade | Oil content volume, percent |
|---|---|
| 1 and 2 | 20 |
| 3 and 4 | 19 |

3.4 Mechanical properties.

3.4.1 Radial crushing strength. Radial crushing strength shall not be less than the value calculated as follows:

$$P = KLT^2/(D - T)$$

where: P = radial crushing strength, lb.
D = outside diameter of bearing, in.
T = wall thickness of bearing, in.
K = strength constant as shown in table IV for grade specified
L = length of bearing, in.

Table IV. Strength constants

| Grade | K (strength constant) |
|---|---|
| 1 | 25,000 |
| 2 | 30,000 |
| 3 and 4 | 40,000 |

3.5 Permissible variations in dimensions. Permissible variations in finished dimensions shall be within the limits specified on drawing or purchase order. Unfinished bars shall be supplied with sufficient stock in length, outside diameter and inside diameter (if applicable) to permit a machine cut of 1/32 to 1/16 inches on each surface to meet nominal specified dimensions.

3.6 Workmanship. Bars shall be uniform in composition. When cut or fractured, the exposed surface shall be of uniform appearance. The bars shall be free from defects which would affect their serviceability.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Lot. A lot shall consist of all bars of the same dimensions made from metal powders of the same type and compositions, formed and sintered under the same conditions, and submitted for inspection at the same time and place.

4.3 Examination.

4.3.1 Visual and dimensional examination. All bars shall be subjected to visual and dimensional examination.

4.4 Sampling.

4.4.1 For chemical analysis. A minimum of one bar shall be taken from each lot for chemical analysis.

4.4.2 For physical properties. Samples for inspection shall be in accordance with MIL-STD-105, level I, unless otherwise specified, the acceptable quality level shall be 1.5 percent.

4.4.3 For strength. A minimum of one sample for radial crushing strength shall be taken from each lot (see 4.6.1.7).

4.5 Examination.

4.5.1 Dimensional. Each bar in the lot shall be examined for compliance with size and tolerance requirements as applicable (see 3.5).

4.5.2 Visual. Each bar in the lot shall be examined for compliance with the requirements for finish and workmanship (see 3.6).



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4.5.3 Preparation for shipment. Examination of the packaging, packing and marking for shipment shall be made for conformance to the requirements of section 5.

4.6 Tests.

4.6.1 Test specimens.

4.6.1.1 Chemical composition. Specimens shall consist of not less than two (2) ounches of chips obtained by milling or drilling with clean dry tools without lubrication. In order to obtain oil free chips, the parts selected for test shall have the lubricant extracted in accordance with 4.6.1.2 or the chips shall be treated with petroleum ether until all of the oil is removed.

4.6.1.2 Void by volume. Bars to be used as specimens shall have the lubricant extracted as follows: The lubricant shall be extracted by heating specimen in a controlled atmosphere (dissociated ammonia or pure hydrogen) furnace at 1400-1600°F for 15 minutes and then cooling to room temperature in the same atmosphere. The alternate method in 4.6.1.3 may be used.

4.6.1.3 Alternate method. Bars to be used as specimens shall have the lubricant extracted in a Soxhlet apparatus of suitable size using toluol or petroleum ether as a solvent. After extraction, residual solvent shall be removed by heating the specimens in an oven for one hour at 250°F (120°C). Alternate extraction and drying shall be continued until the dry weight is constant to 0.1 percent.

4.6.1.4 Strength. A cylindrical (bearing) specimen shall be machined from the sample selected in accordance with 4.4.3 for the purpose of testing radial crushing load specified in 3.4.1. The size of the specimen shall be the nominal dimensions specified under 6.2 with a standard length of 1 inch. In the case of solid bar stock, the nominal O.D. specified will apply with a standard wall thickness of 1/8 inch.

4.6.1.5 Density and porosity. ASTM method B 328 shall be employed to determine these properties.

4.6.1.6 Oil excretion. The bars shall be heated in an oven or other suitable means to a temperature not exceeding 300°F (149°C) and held for a period of not more than 5 minutes. During this period, an oil film or beads shall exude uniformly from all surfaces.



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4.6.1.7 Strength (radial crushing load). The cylindrical bearings shall be compressed radially by a gradually applied load until the load drops owing to the first crack. The cylindrical bearings shall meet the minimum strength requirements as computed from the formula in 3.4.1.

4.7 Rejection. When failure of a sample is definitely acribed to faulty material, the entire lot of material represented shall be rejected subject to the provisions of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. Sintered bars shall be packaged in accordance with method III of MIL-P-116. Each bar shall be wrapped in oil resistant paper.

5.1.2 Level C. Sintered bars shall be packaged in containers of the type commonly used for the purpose, in accordance with commercial practices..

5.2 Packing.

5.2.1 Level A. Bars packaged as specified in 5.1 shall be packed for shipment in containers conforming to any of the following specifications:

PPP-B-636 - Fiberboard - Class 2
PPP-B-576 - Wood-Cleated, Veneer Paper Overlaid, Class 2
PPP-B-585 - Wood-Wirebound - Class 2
PPP-B-591 - Fiberboard, Wood-Cleated, Style A or B
PPP-B-601 - Wood-Cleated Plywood - Overseas Type
PPP-B-621 - Wood-Nailed and Lock Corner - Class 2
PPP-B-640 - Fiberboard, Triple Wall - Class 2

5.2.2 Level B. Bars packaged as specified in 5.1 shall be packed as specified in 5.2.1 except that domestic type containers shall be used.

5.2.3 Level C. Bars shall be packed in containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier. Shipping containers shall comply with the uniform freight classification rules, or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. All interior and exterior containers shall be marked in accordance with MIL-STD-129.



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## 6. NOTES

6.1 Intended use. The bars covered by this specification are intended for use in the manufacture of bearings, bushings, spacers, etc. that are self-lubricated, and that must withstand high shock and impact loads.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

a. Title, number and date of this specification.
b. Grade and form of material (see 1.2 and 1.3).
c. Oil to be used for impregnation (see 3.3.1).
d. Sampling for physical properties (see 4.4.2).
e. Size required (see 3.5).
f. Level of packaging and packing (see 5.1 and 5.2).

Custodian:
Army - MU
Navy - SH
Air Force - 69

Review interest:
Army - MU, MR
Navy - SH
Air Force - 69

User interest:
Army - GL, ME, MI

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